Harrison Casper

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EDUCATION

Brigham Young University, B.S. Computer Science — Aug 2020 – Apr 2026
GPA: 3.9 | Coursework: Machine Learning, Database Modeling Concepts, Software Engineering, Discrete Structures, DevOps
Teaching Assistant, *BYU Computer Science* — Jan 2024 – Apr 2026
CS 236 Discrete Structures (4 semesters, Winter 2024 – Summer 2025; 200+ students, led weekly labs)
CS 452 Database Modeling Concepts (Spring 2025); CS 393 Collaborative Problem Solving (Fall 2025 – Winter 2026)

EXPERIENCE

Spout Inc. | Software Developer Intern — May 2026 – Aug 2026
- Shipped full-stack features across a Java backend and Angular/TypeScript frontend, including a new API endpoint and its typed client; fixed backend sorting and status logic.
- Wrote and updated Java tests (PostgreSQL integration, service, controller) and Angular spec tests.
- Built an end-to-end testing system from scratch with Playwright that runs post-merge in GitHub Actions against an ephemeral seeded test environment, with automatic Teams alerts on failure.
- Built new features including a mailing address manager and Mapbox map zoning.
- Prototyped an iOS and Android mobile app in NativeScript; cleaned up and restructured the code repository.

LocalFiber | UX/UI Design Intern — Feb 2026 – May 2026
- Designed and built a React and Tailwind CSS dashboard showing customer and advertising data for ISPs nationwide.
- Integrated the dashboard's React components with REST API endpoints and styled the interface with Tailwind CSS.

EduQuest (AI ed-tech startup) | Research Assistant — Sep 2025 – Dec 2025
- Co-supervised a DynamoDB redesign with one coworker; rewrote data models in Python/boto3, replacing embedded lists with ID references and full-table scans with indexed queries.
- Built a table-cloning utility for isolated test tables and wrote ~1,100 lines of pytest tests for the quest data layer.
- Parallelized quest generation with asyncio; shipped work through the team's Azure DevOps workflow (Boards, Pipelines).

Brigham Young University, Astronomy Research | Research Assistant — Apr 2025 – Aug 2025
- Designed and trained a PyTorch CNN from scratch to classify galaxy images as the team's only ML contributor.

PROJECTS

booktrakd | TypeScript, Next.js, tRPC, Drizzle ORM, PostgreSQL, Vitest — Jan 2026 – Present
- Built a full-stack social site to log, rate, and review books, with follows and an activity feed; deployed on Vercel + Neon.
- Wrote a Node.js import pipeline and job worker that dedupes Open Library data into a ~49,000-book, 800,000-edition catalog; added Vitest tests and GitHub Actions CI.

General Conference RAG | BYU CS 452 TA project | Python, OpenAI API, Sentence Transformers — May 2025 – Jun 2025
- Built from scratch as a TA, at a professor's request, a retrieval-augmented ChatGPT system over LDS General Conference talks (scraping, embeddings, grounded answers), plus a course assignment using it as student starter code.

JWT Pizza | BYU DevOps course project | Playwright, Jest, Docker, GitHub Actions, AWS — Sep 2024 – Dec 2024
- Wrote Playwright end-to-end tests for a React app (checkout, registration, admin, navigation) with mocked API routes.
- Brought an Express/MySQL API to ~79% test coverage (Jest/Supertest); added Docker and GitHub Actions CI/CD to AWS.

Elementary (incremental game) | JavaScript/TypeScript, React, Svelte, PixiJS, Express, AWS — Jan 2024 – Present
- Built the original solo for BYU CS 260 (2024): element-fusion mechanics in React, plus Express, WebSockets, and bcrypt/MongoDB auth on AWS EC2; now directing an AI-assisted rebuild in TypeScript, Svelte 5, PixiJS, and Capacitor.

SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML/CSS, C++ (coursework)
Frontend: React, Angular, PrimeNG, RxJS, Next.js, Svelte, PixiJS, Tailwind CSS, Vite, Mapbox, NativeScript (iOS/Android)
Backend & Databases: Node.js, Express, REST APIs, tRPC, WebSockets, Maven, PostgreSQL, MySQL, MongoDB, DynamoDB
Testing: Playwright, JUnit, Jest, Supertest, Vitest, pytest, Angular unit tests, end-to-end and integration testing
DevOps & Tools: Git/GitHub, GitHub Actions, Docker, AWS (ECR, ECS, CloudFormation, S3), Azure DevOps, Vercel, pnpm
AI/ML: PyTorch, CNNs, RAG, OpenAI API, Sentence Transformers, AI coding agents